

Linux Learning Guide

Resources, Projects & Daily Practice for Beginners

This guide is designed for absolute beginners who have just started learning Linux. You'll find curated websites, daily project ideas, and a step-by-step learning path to help you build real skills through consistent practice.

■ 6 Top Resources

■■ Daily Practice Plan

■■ 10+ Project Ideas

■■ Clear Learning Path

1. Interactive Practice Platforms

These platforms give you a real or simulated Linux terminal in your browser — no installation needed. Perfect for daily 15–20 minute sessions.

OverTheWire: Bandit

● Beginner

overthewire.org/wargames/bandit

Wargames that teach Linux through fun, hacking-style challenges. Each level requires you to use Linux commands to find a password and move to the next level. Start at Level 0 — it's perfect right after learning `ls`.

■ *Best starting point after learning `ls`, `cd`, `cat`, and `file`.*

Linux Survival

● Beginner

linuxsurvival.com

A browser-based Linux terminal simulator with step-by-step guided lessons. You type real commands and get instant feedback. Covers navigation, file management, permissions, and more.

■ *Use this before OverTheWire — it's gentler and more guided.*

KillerCoda

● Beginner–Intermediate

killercoda.com

Provides real Linux environments (Ubuntu) in your browser with structured scenarios. Has dedicated beginner Linux tracks and also covers Docker, Kubernetes, and DevOps tools as you grow.

■ *Try the 'Ubuntu Playground' for free-form exploration.*

TryHackMe

● Beginner

tryhackme.com

A gamified cybersecurity and Linux learning platform. The 'Linux Fundamentals' path (Parts 1–3) is an excellent structured beginner course. Earn points and badges as you progress.

■ *Complete 'Linux Fundamentals Part 1' as your first structured course.*

CommandChallenge

● All Levels

cmdchallenge.com

One Linux command challenge at a time, solved directly in the browser. Challenges range from super simple (list files) to advanced (one-liners with pipes). Great for a quick daily challenge.

■ *Do 1–2 challenges per day as a warm-up habit.*

Linux Journey

● Beginner

linuxjourney.com

A beautifully designed, free website that teaches Linux concepts with short lessons and quizzes. Covers Grasshopper (beginner) through Journeyman topics. Read a lesson, then practice it on the terminal.

■ *Read one lesson here, then practice the commands on KillerCoda.*

2. Daily Practice Plan (15–30 Minutes)

Day / Focus	Activity	Platform
Monday Navigation	Practice ls, cd, pwd, mkdir, rmdir Explore directory trees	Linux Survival
Tuesday Files	Use cp, mv, rm, touch, cat, less Create & organize files	KillerCoda Playground
Wednesday Challenge	Solve 2–3 command challenges Focus on what you learnt this week	CommandChallenge
Thursday Permissions	Study chmod, chown, ls -l Understand rwx flags	Linux Journey
Friday Text Tools	Use grep, find, sort, head, tail Search inside files	OverTheWire Bandit
Saturday Project	Build something small (see Section 3) Apply all week's commands	KillerCoda / Local
Sunday Review	Re-read notes, redo tricky challenges Write down 5 things you learned	Any

3. Beginner Project Ideas

Each project below uses only basic Linux commands you'll learn in your first few weeks. Try one per week after you've practiced the relevant commands.

Week 1–2 File System Explorer

Using only ls, cd, mkdir, touch, and pwd — build a folder structure for a fake school: school/ → subjects/ → math/, science/, english/ Inside each: notes.txt, homework.txt

Week 2–3 Personal Journal in Terminal

Use echo, cat, and >> to write daily journal entries into a .txt file. Use grep to search entries by keyword. Practice date >> journal.txt to timestamp entries.

Week 3–4 File Organizer

Create a folder with mixed files (photos, docs, music). Use mv and mkdir to sort them into organized subfolders by type. Practice ls -lh to verify sizes.

Week 4–5	System Info Reporter
Use commands like <code>uname -a</code> , <code>df -h</code> , <code>free -h</code> , <code>uptime</code> , <code>whoami</code> , and <code>hostname</code> . Pipe output with <code>></code> to save a <code>system_report.txt</code> file.	
Month 2	Simple Bash Script
Write your first shell script: a <code>backup.sh</code> that copies important files to a <code>backup/</code> folder using <code>cp -r</code> . Run it with <code>bash backup.sh</code> .	

4. Your Step-by-Step Learning Path

Stage	Commands to Learn	Goal
■ Stage 1 Absolute Beginner (Week 1–2)	<code>ls</code> , <code>cd</code> , <code>pwd</code> <code>mkdir</code> , <code>rmdir</code> , <code>touch</code> <code>cat</code> , <code>less</code> , <code>more</code>	Navigate the file system confidently. Know where you are at all times.
■ Stage 2 File Management (Week 2–4)	<code>cp</code> , <code>mv</code> , <code>rm</code> <code>find</code> , <code>locate</code> <code>head</code> , <code>tail</code> , <code>wc</code>	Create, move, copy and delete files. Find files anywhere on the system.
■ Stage 3 Text & Search (Week 4–6)	<code>grep</code> , <code>sort</code> , <code>uniq</code> <code>cut</code> , <code>awk</code> , <code>sed</code> <code>echo</code> , <code>printf</code>	Search inside files and filter output. Manipulate text from the terminal.
■ Stage 4 Permissions & Users (Week 6–8)	<code>chmod</code> , <code>chown</code> <code>sudo</code> , <code>su</code> <code>whoami</code> , <code>groups</code>	Understand Linux permissions. Run commands as root safely.
■ Stage 5 Processes & System (Week 8–10)	<code>ps</code> , <code>top</code> , <code>kill</code> <code>df</code> , <code>du</code> , <code>free</code> <code>uname</code> , <code>uptime</code>	Monitor system resources. Manage running processes.
■ Stage 6 Scripting Basics (Month 3+)	bash scripts variables, loops <code>if/else</code> , functions	Write your first shell scripts to automate repetitive tasks.

5. Tips for Consistent Learning

■ ■ Be consistent, not intense	15–20 minutes every day beats 3 hours once a week. Use CommandChallenge as your daily warm-up.
■ Keep a command notebook	Write down every new command with an example. Review it weekly. This builds muscle memory faster.
■ Embrace errors	When a command fails, read the error message carefully. Linux errors are very descriptive and teach you a lot.
■ Use man pages	Type <code>man ls</code> or <code>man grep</code> to read the official manual for any command. Use <code>/keyword</code> to search inside.

■ **Repeat, repeat, repeat**

Re-do exercises you found hard. Mastery comes from repetition, not just exposure.

■ **Focus on one concept per day**

Don't try to learn 20 commands at once. Learn 2–3 deeply and practice them in real contexts.

Created for Linux beginners · Start small, stay consistent, and you'll be surprised how fast you grow ■